

SQL Homework 2: MySQL Employees Database

Topic: Queries Using MySQL Employees Database

Instructions:

•Use the MySQL Employees Database (schema: employees, salaries, departments, dept_emp, dept_manager, titles).

•Write the SQL queries for each question and include the results as part of your submission.

•Ensure your queries include proper formatting, comments, and structure.

•Submission Deadline: 11th January 2025 17:00

Part 1: Advanced Joins (1-5)

- 1.Retrieve the **first name**, **last name**, and **department name** of employees who started working **after their department manager's hire date**.
- 2.List the **department names** and the total **number of employees** in each department who have the title of "Senior Engineer".
- 3.Find the **first name**, **last name**, and **salary** of employees who have worked in **multiple departments**.
- 4.Retrieve the **employee numbers** of employees who currently do not have any active title assignments.
- 5.List all departments and the **number of employees** in each department who earn more than **100,000**.

Part 2: Nested Queries (6-10)

- 6.Find the **first name** and **last name** of employees who earn more than the **highest salary** in department "d003".
- 7.Retrieve the **employee numbers** of employees whose salaries are greater than the **average salary** of all employees in the same department.
- 8.Find the **departments** where at least one employee earns more than **80,000** and works as a "Staff".
- 9.Retrieve the **employee numbers** of employees who have never earned more than **50,000** during their entire career.
- 10.Find the **department names** where all employees earn less than **90,000**.

Part 3: Aggregations (11-15)

- 11.Find the **average salary** of employees for each title.
- 12.List the **top 3 departments** with the **highest average salaries** across all employees.
- 13.Retrieve the **employee numbers** and the **total number of titles** they have held over their career.
- 14.Find the **department name** and the **total compensation** of employees for each department.
- 15.Retrieve the **first name**, **last name**, and **maximum salary** of employees who have been "Managers"

Part 4: Set Operations (16-20)

16. Retrieve the **names** of employees who have worked in both “Development” and “Sales” departments.
17. List the **employee numbers** of employees who have had the title “Assistant Engineer” but have **never worked** in the “Production” department.
18. Retrieve the **department names** where no employees currently hold the title “Technician”.
19. Find the **names** of employees who have worked as both “Senior Engineer” and “Senior Staff”.
20. List all **departments** and indicate whether each has at least one employee earning more than **150,000** (“Yes” or “No”).

Submission Instructions:

1. Write all 20 queries in a .sql file with proper comments indicating the question number.
2. Run each query on the MySQL Employees Database and include the query output (screenshots or exported results).
3. Ensure your queries are formatted for readability and correctness.